complete java class code example of each combination of intermediate and terminal operations

filter() boolean checks + allmatch() validation logic

filter() null checks + allmatch() validation logic

filter() comparisons + allmatch() validation logic

filter() method calls returning boolean + allmatch() validation logic

distinct() rely on equals() / hashcode() + allmatch() validation logic

distinct() immutable objects + allmatch() validation logic

map() transform objects + allmatch() validation logic

map() create new objects + allmatch() validation logic

map() call pure functions + allmatch() validation logic

flatmap() return stream + allmatch() validation logic

flatmap() split data + allmatch() validation logic

flatmap() expand collections + allmatch() validation logic

sorted() natural ordering + allmatch() validation logic

sorted() custom comparator + allmatch() validation logic

sorted() multi-field compare + allmatch() validation logic

peek() logging + allmatch() validation logic

peek() debugging + allmatch() validation logic

peek() metrics + allmatch() validation logic

Below are **all 18 programs converted to Java 1.5 compatible code**.

Since Java 1.5:

- ❌ No Stream API
- ❌ No Optional
- ❌ No lambda expressions
- ❌ No method references
- ❌ No List.of(), Map.of()
- ❌ No Objects.hash()
- ❌ No Comparator.comparing()

We replace everything with:

- ✔ for-each loops
- ✔ Iterator
- ✔ Collections / Arrays
- ✔ HashSet for distinct
- ✔ Manual sorting with Comparator
- ✔ Null checks
- ✔ Primitive counters instead of AtomicInteger

Below are **complete, standalone Java class examples** for **each requested Stream combination**, where:

- **Terminal operation = allMatch()**
- Purpose = **validation logic** (returns true only if **all elements** satisfy the predicate)
- Demonstrates **short-circuiting** (stops on first failure)
- Java **8+** compatible
- No collectors / reduce / custom aggregation

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1 filter() boolean checks + allMatch() (Validation)

```
import java.util.*;

public class FilterBooleanAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of(2, 4, 6, 8).stream()
            .filter(n -> n > 0)
            .allMatch(n -> n % 2 == 0);

      System.out.println(valid);
   }
}
```

---

2 filter() null checks + allMatch() (Validation)

```
import java.util.*;

public class FilterNullAllMatch {
   public static void main(String[] args) {
      boolean valid = Arrays.asList("Anurag", "Amit").stream()
            .filter(Objects::nonNull)
            .allMatch(s -> !s.isEmpty());

      System.out.println(valid);
   }
}
```

---

3 filter() comparisons + allMatch() (Validation)

```
import java.util.*;

public class FilterComparisonAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of(30, 40, 50).stream()
            .filter(age -> age >= 18)
            .allMatch(age -> age <= 60);

      System.out.println(valid);
   }
}
```

---

4 filter() method calls returning boolean + allMatch() (Validation)

```
import java.util.*;

public class FilterMethodAllMatch {

   static class User {
      boolean active;
      User(boolean active) { this.active = active; }
      boolean isActive() { return active; }
   }

   public static void main(String[] args) {
      boolean valid = List.of(
            new User(true),
            new User(true)
      ).stream()
       .filter(User::isActive)
       .allMatch(User::isActive);

      System.out.println(valid);
   }
}
```

---

5 distinct() rely on equals() / hashCode() + allMatch() (Validation)

```
import java.util.*;

public class DistinctEqualsAllMatch {

   static class Person {
      String name;
      Person(String name) { this.name = name; }

      @Override public boolean equals(Object o) {
         return o instanceof Person p && name.equals(p.name);
      }
      @Override public int hashCode() {
         return Objects.hash(name);
      }
   }

   public static void main(String[] args) {
      boolean valid = List.of(
            new Person("Anurag"),
            new Person("Anurag"),
            new Person("Ravi")
      ).stream()
       .distinct()
       .allMatch(p -> p.name.length() >= 4);

      System.out.println(valid);
   }
}
```

---

6 distinct() immutable objects + allMatch() (Validation)

```
import java.util.*;

public class DistinctImmutableAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of("Java", "Java", "Go").stream()
            .distinct()
            .allMatch(s -> s.length() >= 2);

      System.out.println(valid);
   }
}
```

---

7 map() transform objects + allMatch() (Validation)

```
import java.util.*;

public class MapTransformAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of("anurag", "amit").stream()
            .map(String::toUpperCase)
            .allMatch(s -> s.equals(s.toUpperCase()));

      System.out.println(valid);
   }
}
```

---

8 map() create new objects + allMatch() (Validation)

```
import java.util.*;

public class MapCreateObjectAllMatch {

   static class UserDTO {
      String name;
      UserDTO(String name) { this.name = name; }
   }

   public static void main(String[] args) {
      boolean valid = List.of("Anurag", "Amit").stream()
            .map(UserDTO::new)
            .allMatch(dto -> dto.name.length() >= 4);

      System.out.println(valid);
   }
}
```

---

9 map() call pure functions + allMatch() (Validation)

```
import java.util.*;

public class MapPureFunctionAllMatch {

   static int square(int n) { return n * n; }

   public static void main(String[] args) {
      boolean valid = List.of(2, 3, 4).stream()
            .map(MapPureFunctionAllMatch::square)
            .allMatch(n -> n > 0);

      System.out.println(valid);
   }
}
```

---

10 flatMap() return Stream + allMatch() (Validation)

```
import java.util.*;

public class FlatMapStreamAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of(
            List.of(1, 2),
            List.of(3, 4)
      ).stream()
       .flatMap(List::stream)
       .allMatch(n -> n > 0);

      System.out.println(valid);
   }
}
```

---

11 flatMap() split data + allMatch() (Validation)

```
import java.util.*;

public class FlatMapSplitAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of("Java Stream", "Flat Map").stream()
            .flatMap(s -> Arrays.stream(s.split(" ")))
            .allMatch(word -> word.length() >= 3);

      System.out.println(valid);
   }
}
```

---

12 flatMap() expand collections + allMatch() (Validation)

```
import java.util.*;

public class FlatMapExpandAllMatch {
   public static void main(String[] args) {
      boolean valid = Map.of(
            "A", List.of("1", "2"),
            "B", List.of("3")
      ).values().stream()
       .flatMap(Collection::stream)
       .allMatch(v -> !v.isEmpty());

      System.out.println(valid);
   }
}
```

---

13 sorted() natural ordering + allMatch() (Validation)

```
import java.util.*;

public class SortedNaturalAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of(1, 2, 3).stream()
            .sorted()
            .allMatch(n -> n >= 1);

      System.out.println(valid);
   }
}
```

---

14 sorted() custom comparator + allMatch() (Validation)

```
import java.util.*;

public class SortedCustomAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of("Anurag", "Ravi", "Amit").stream()
            .sorted(Comparator.reverseOrder())
            .allMatch(name -> name.length() >= 4);

      System.out.println(valid);
   }
}
```

---

15 sorted() multi-field compare + allMatch() (Validation)

```
import java.util.*;

public class SortedMultiFieldAllMatch {

   static class Employee {
      String name;
      int age;
      Employee(String name, int age) {
         this.name = name; this.age = age;
      }
   }

   public static void main(String[] args) {
      boolean valid = List.of(
            new Employee("Anurag", 30),
            new Employee("Anurag", 25),
            new Employee("Ravi", 35)
      ).stream()
       .sorted(Comparator
             .comparing((Employee e) -> e.name)
             .thenComparingInt(e -> e.age))
       .allMatch(e -> e.age >= 18);

      System.out.println(valid);
   }
}
```

---

16 peek() logging + allMatch() (Validation)

```
import java.util.*;

public class PeekLoggingAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of(1, 2, 3).stream()
            .peek(n -> System.out.println("Checking: " + n))
            .allMatch(n -> n > 0);

      System.out.println(valid);
   }
}
```

---

17 peek() debugging + allMatch() (Validation)

```
import java.util.*;

public class PeekDebugAllMatch {
   public static void main(String[] args) {
      boolean valid = List.of(10, 20, 30).stream()
            .peek(n -> {
               if (n == 20) System.out.println("Debug hit");
            })
            .allMatch(n -> n >= 10);

      System.out.println(valid);
   }
}
```

---

18 peek() metrics + allMatch() (Validation)

```
import java.util.*;
import java.util.concurrent.atomic.AtomicInteger;

public class PeekMetricsAllMatch {
   public static void main(String[] args) {
      AtomicInteger counter = new AtomicInteger();

      boolean valid = List.of(5, 10, 15).stream()
            .peek(n -> counter.incrementAndGet())
            .allMatch(n -> n > 0);

      System.out.println("Valid: " + valid);
      System.out.println("Processed elements: " + counter.get());
   }
}
```